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A data-driven approach to predicting power outages during winter storms in the southern U.S. leveraging nonparametric machine learning models

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Abstract

In February 2021, Winter Storm Uri severely impacted much of the southern United States, triggering unprecedented large-scale power outages. Recognizing that a similar extreme weather event could occur in the future, this study identifies as its primary research objective the development of a baseline power outage prediction model specifically tailored for the southern region of the United States. Central to this objective is the research question: Which variables and which regression models play the most significant role in accurately predicting power outages in this context? Given that large-scale outages are, in essence, a direct result of imbalances between electricity supply and demand, population was considered a key influencing factor. Furthermore, to ensure the model adequately reflects the meteorological characteristics of winter storms, several atmospheric variables—such as dew point and atmospheric pressure—were incorporated into the analysis. These variables are intended to capture the environmental dynamics that underpin outage occurrence during extreme cold events. Four machine learning models—Random Forest, eXtreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LightGBM), and Categorical Boosting (CatBoost)—were employed in this study. In addition, to enable a comparison between these four machine learning approaches and traditional statistical models, Ridge regression and Lasso regression were also implemented, utilizing population and geographic information data in conjunction with meteorological variables to achieve this objective. To determine the optimal model configuration, Bayesian optimization was employed using tenfold cross-validation. The results revealed that XGBoost achieved the highest performance, with an R^2 score of 0.92. Furthermore, when the XGBoost model was utilized for prediction, a permutation importance analysis identified population, dew point, and pressure—in that order—as the most influential variables. Additionally, given that the number of data points varied by state during the test evaluation phase, a weighted evaluation metric was also computed using the data counts for each state. Under this weighted evaluation, XGBoost still achieved the highest R^2 score (0.74), further underscoring its robustness across heterogeneous state-level datasets. Consequently, this paper developed a foundational baseline model for power outage prediction due to winter storms in the Southern United States and identified essential variables through analysis.

Keywords Power outages, Data-Driven, Winter Storms, Machine learning

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1 Introduction

1.1 Research background and motivation

Global warming and climate change have made extreme weather phenomena such as floods, droughts, and severe storms more frequent (Hoegh-Guldberg et al., 2018; Yadollahie, 2019; Zandalinas et al., 2021). These events cause physical and social disruption, and as the risks increase, communities need to respond more quickly and effectively to volatile climate conditions.

A representative example is the unprecedented winter storm "Uri," which landed the southern United States in February 2021, causing over $195 billion in damages (The City of Austin & Travis County, 2022). The storm triggered massive power outages across the country, from the Southwest to the Northeast. Northern states (such as Illinois and New York) are relatively well-prepared and experienced in dealing with winter storms, whereas southern states (such as Texas) are highly vulnerable to extreme cold weather, resulting in severe damage. Consequently, the southern region experienced unprecedented power supply problems and significant social hardship.

The direct cause of the power outages during Winter Storm Uri was found to be an imbalance between electricity supply and demand. However, if an appropriate outage prediction model had been developed, it could have been used to identify priority areas for electricity supply and thereby minimize the damage. For example, if the prediction model indicated that Region A would have more outages than Region B within the next few hours, the crisis response team could prioritize electricity supply to Region A or implement rolling blackouts to mitigate the impact. This simplified example demonstrates the usefulness of having an accurate model during winter storm events. In order to develop such an outage prediction model, it is essential to examine the extent of existing research.

Existing research on power outage prediction has mainly concentrated on hurricanes (Arora & Ceferino, 2023, 2024; Guikema & Quiring, 2012; Guikema et al., 2006, 2010, 2014); Han et al., 2009a, 2009b; Liu et al., 2005; McRoberts et al., 2018; Nateghi et al., 2014; Quiring et al., 2011; Shashaani et al., 2018; Tonn et al., 2016; Wanik et al., 2015, 2017; Yang et al., 2020), whereas research specifically addressing winter storms is comparatively scarce.

Hurricane-related outage prediction studies have employed both parametric and nonparametric modeling approaches. While parametric models rely on a limited set of parameters and assume specific probability distributions, nonparametric models operate without such assumptions, thereby providing greater flexibility in capturing complex data characteristics. Considering these advantages, the present study adopts a nonparametric regression approach.

Research on winter storms (Bhattacharyya & Hastak, 2022; Grineski et al., 2022, 2023; Lee et al., 2022; Lin et al., 2021; Melaku et al., 2023; Nejat et al., 2022) has predominantly examined socio-economic effects and community resilience, leaving a notable gap in the development of predictive models for winter storm-related power outages in the United States. Addressing this gap, several prior studies have laid important groundwork, as discussed below.

Cerrai et al. (2020) developed power outage prediction models for 54 snow and ice storm events in Connecticut using a variety of variables such as meteorological data, terrain, infrastructure, and vegetation information. The Random Forest model and the Bayesian Additive Regression Trees (Random Forest, Bayesian Additive Regression Trees) were effective at predicting recurring, low-impact events, while the Generalized Linear Model (Generalized Linear Model, GLM) was strong at predicting extreme events. Snow density and leaf area index were identified as significant variables.

Allen-Dumas et al. (2022) predicted outages and conducted correlation analysis using the Random Forest regression model (Random Forest regression) with county-level and 1-km grid meteorological and outage data during the 2021 Uri and 2022 Landon winter storms in Texas. The results showed that the previous day's outage rate was a strong predictor for the next day's rate, and prediction accuracy improved as the dataset was expanded.

Lee et al. (2023) combined power outage data from multiple U.S. states (Texas, Michigan, Hawaii) with weather information from the National Weather Service (National Weather Service, NWS) to develop state-level machine learning models (Random Forest, k-Nearest Neighbor regression, Extreme Gradient Boosting; Random Forest, k-NN, XGBoost), achieving over 90% predictive accuracy. As more historical data was available, model performance improved, and future research is expected to refine predictions at the county level.

In summary, research on power outage prediction for winter storms is still limited compared to that for hurricanes. Incidents like the 2021 Texas Winter storm Uri crisis underscore the importance of identifying necessary variables, algorithms, and integrated data. The studies by Cerrai et al. (2020), Allen-Dumas et al. (2022), and Lee et al. (2023) are pioneering works and serve as key references for this research topic.

Although the study by Cerrai et al. (2020) focused on snow and ice storms in the northern United States, the meteorological and infrastructure impact mechanisms observed in the north (e.g., snow loading) are substantially different from those winter storms in the southern regions (e.g., sudden cold waves causing imbalances in electricity supply and demand). Therefore, the works by Lee et al. (2023) and Allen-Dumas et al. (2022) are more closely related to the present research topic.

For this reason, these two studies will be discussed in more detail in the related works section. Overall, while outage prediction models for hurricanes have advanced considerably, studies on winter storms remain relatively scarce. Integrating a wide range of indicators beyond meteorological factors and evaluating both parametric and nonparametric prediction methodologies are essential for developing more reliable outage prediction models for winter storms.

1.2 Related works

In the last five years, research on power outages or failures caused by winter storms in southern areas has been limited, mainly because detailed, high-quality outage information is difficult to obtain given the low frequency of winter storms and the lack of direct access to utility company records. Consequently, a large amount of research carried out up to now has focused on outage prediction models for hurricanes, not because hurricanes are naturally more frequent than winter storms, but because they produce immediate, intense, and highly visible impacts, especially through destructive winds, making them simpler to analyze for both researchers and utilities.

Conversely, winter storms pose more complex difficulties for prediction. The strength of winter storms is not as distinctly categorized as that of hurricanes, making it more difficult to differentiate between mild and extreme winter occurrences. While wind plays a role, outages during winter storms are often due to other factors, such as ice accumulation damaging infrastructure, different types of precipitation, or sudden drops in temperature (Allen-Dumas et al., 2022). Despite several recent efforts to model these effects, the overall body of literature remains limited in scope and resolution (Allen-Dumas et al., 2022).

Recent studies have begun to connect this divide. Allen-Dumas et al. (2022) analyzed the correlation between various cold-weather meteorological variables—such as maximum and minimum temperature, precipitation, and snow accumulation—and power outage occurrences during the "Winter Storm Uri" in Texas in February 2021 and the "Winter Storm Landon" in February 2022.

The purpose of Allen-Dumas et al. (2022)'s study was to use big data and machine learning techniques (specifically, the random forest model) with Pearson's correlation coefficient to determine whether today's and yesterday's meteorological information can be used to predict tomorrow's power outages. The ultimate goal was to improve grid reliability in preparation for extreme weather events such as cold waves.

The data used in the study included meteorological observations at a one-kilometer by one-kilometer grid scale (maximum/minimum temperature, cumulative precipitation, daily average incident shortwave radiation, snow water equivalent, average vapor pressure, etc.) and daily county-level electricity outage reports in Texas (based on the number of affected customers). These datasets were collected from the Daymet Version 3 dataset at Oak Ridge National Laboratory and the Environment for Analysis of Geo-Located Energy Information (EAGLE-I) platform maintained by the United States Department of Energy (DOE). To achieve the study's objectives, the authors seek to predict next-day outage occurrences (outage customer ratio, predicted percentage of customers without power) using meteorological data and previous-day outage data.

The independent variables included today's and yesterday's maximum and minimum temperatures, precipitation, snow water equivalent, incident shortwave radiation, vapor pressure, population count, electricity customer count, and previous outage values—covering various meteorological and electricity-related factors. The dependent variable was defined as the next day's county-level percentage of customers experiencing power outages. The results indicated that for both storms, the previous day's outage ratio was the most highly positively correlated variable with the next day's outage ratio. Meteorological variables alone did not exhibit a clear or consistent strong correlation for predicting power outages. The predictive performance improved slightly with more data (e.g., merging data from both storm events), and Allen-Dumas et al. (2022) concluded that non-linear patterns might exist that cannot be captured solely by Pearson's correlation coefficient.

Allen-Dumas et al. (2022) identified four key limitations in their study. First, the use of only Pearson's correlation coefficient meant that more complex, non-linear correlation structures could not be captured. Second, external factors such as infrastructure differences, preparedness levels in different counties, and incomplete or inaccurate data (e.g., over- or under-reporting electricity customers) could have affected the results. Third, expanding the analysis to cover other regions, years, or storms is necessary. Lastly, issues such as the treatment of missing data, data cleaning processes, and inconsistencies in data availability were identified as further limitations.

Lee et al. (2023) developed a machine learning model to predict power outages caused by extreme weather events such as hurricanes, storms, and floods. Their research predicted the number of power outages at the state level across the United States during extreme weather events, enabling proactive response and recovery planning, and assisting utility companies and emergency management agencies in responding more efficiently.

The data they used combined historic U.S. power outage data collected by the Environment for Analysis of Geo-Located Energy Information (EAGLE-I) system and weather alert data provided by the U.S. National Weather Service (NWS). The EAGLE-I data includes records of power outage customers in each county at 15-min intervals since 2014. Also, the NWS data contains geographic information, types of weather alerts (such as hurricane, flood, thunderstorm), and alert levels (Watch/Warning/Advisory). They used the above data to ultimately predict, at the state level, the sum of maximum county-level outages in the 12 h following a weather alert event. In other words, they sought to predict "the sum of the maximum number of customers affected by power outages in each county within 12 h after a given weather alert is issued".

Their input features included the maximum outages in each county during the 12 h before the alert, the type of weather alert (such as flood, thunderstorm), and a list of counties affected by the alert. Their prediction (target) variable was the sum of the maximum number of outages in each county for each state during the 12 h after the alert was issued.

They reported that for states with sufficient data (e.g., Texas, Michigan), all three models (Random Forest, k-Nearest Neighbors, Extreme Gradient Boosting) achieved very high predictive performance (R-squared greater than 0.98), and even regions with less data (e.g., Hawaii) performed comparatively well. Additionally, the Extreme Gradient Boosting model performed best in terms of root mean square error (RMSE), and prediction accuracy increased with a higher volume of data per state.

They described four main limitations of their research. First, the amount of data and its regional variance—data amounts differ by region/state, and some areas have few relevant events, making learning difficult. Second, data imbalance—rare weather events lead to a scarcity of records and may decrease predictive accuracy for those areas. Third, the limitation of state-level prediction—future research should seek for more detailed, county-level predictions. Lastly, the limitation of not incorporating complex factors such as infrastructure/network interdependencies within the power grid.

Building on the important foundations laid by prior studies forecasting power outages caused by extreme winter storms and hurricanes, this research offers several distinct contributions that advance the state of the field. Allen-Dumas et al. (2022) primarily focused on predicting the ratio of outages within a limited region, applied only a single machine learning technique, and evaluated variable impacts using more restricted statistical approaches. Similarly, Lee et al. (2023) prioritized state-level outage prediction and compared three public-data-based models, but did not fully incorporate county-level counts of actual outages, deep variable influence analysis, broader model comparisons, or data augmentation strategies.

In contrast, this current study delivers several contributions. First, while conducting an in-depth analysis of a single winter storm, the study broadens the geographic scope to nine states with observed outages, significantly increasing the spatial generalizability and practical relevance of the results. Unlike Allen-Dumas et al. (2022), who predicted only outage percentages, and in line with Lee et al. (2023), this research directly predicts the actual number of outages for each county, maximizing value for policy and operational applications. Going beyond basic correlation analysis, this study employs not only permutation importance but also partial dependence plots and other interpretive tools, greatly enhancing model transparency and reliability. Rather than restricting the comparison to a narrow set of machine learning algorithms, this research systematically compares other machine learning models such as LightGBM and CatBoost as well as classical statistical methods like Lasso regression. Finally, while prior studies (Allen-Dumas et al., 2022; Lee et al., 2023) relied on sufficiently large big data sets, this work applies data augmentation techniques to maximize model generalizability even in rare-event or limited-data situations—despite analyzing just a single event, Winter Storm Uri.

This current research goes beyond the limitations of prior work, providing an accurate, interpretable, and generalizable approach to outage prediction during extreme winter weather events in Southern US.

1.3 Objective

This study focuses on developing a power outage prediction model tailored to the unique characteristics of the southern United States, where winter storms, though rare, can be devastating, as exemplified by Winter Storm Uri in 2021. Since previous literature has primarily modeled winter storm-induced outages based on northern regions of the U.S., it has been difficult to establish a suitable prediction framework for the South; therefore, this study seeks to lay the groundwork for a regionally specialized model.

To achieve this objective, the study poses two key questions: First, which variables most critically affect outage prediction during southern winter storms? Second, which modeling approaches offer the highest predictive performance? By leveraging real observation data and expanding the analysis to nine states that experienced outages, the study strives to provide more interpretable and generalizable results.

In the modeling process, several climate variables, such as dew point and atmospheric pressure, were selected as predictors to reflect the meteorological characteristics of winter storms, while population was included based on the recognition that large-scale outages often stem from supply–demand imbalances. To account for spatial diversity, latitude, longitude, and state identifiers were included, and land area was considered to assess the vulnerability of larger regions. These variables form the logical foundation for the study's modeling and evaluation.

Ultimately, the goal of this research is to identify the key predictors that reflect the distinctive nature of southern US's winter storms and to propose the most appropriate outage prediction modeling approach for real-world operations and policy decisions in regions unaccustomed to extreme cold weather.

2 Dataset description

In February 2021, Winter Storm Uri landed in the United States, primarily affecting nine states— Texas (TX), Arkansas (AR), Louisiana (LA), Mississippi (MS), Tennessee (TN), Kentucky (KY), West Virginia (WV), Virginia (VA), and North Carolina (NC)—as illustrated in Fig. 1.

This winter storm caused widespread power outages, with the state of Texas, particularly vulnerable to extreme winter weather, suffering the most damage. Winter storm Uri had the most significant impact from the 13th to the 15th of February 2021. To achieve a comprehensive understanding of the power outages caused by such winter storms, data from February 11th, 2021 to February 24th, 2021, was utilized. The following subsections will describe the various datasets combined for use in this study.

Prior to providing a detailed description of the various datasets, the geographic, infrastructural, and socioeconomic background of the nine impacted southern U.S. states—Texas, Arkansas, Louisiana, Mississippi, Tennessee, Kentucky, West Virginia, Virginia, and North Carolina—was characterized based on the 2020 U.S. Census.

From a geographical standpoint, the region presents a heterogeneous composition of plains, mountain ranges, coastal zones, major metropolitan centers, and expansive rural areas. Main city areas include Houston, Dallas, and Austin in Texas; Charlotte and Raleigh in North Carolina; and Richmond in Virginia, which serve as principal hubs of population concentration and economic activity. The natural environment is equally diverse, marked by defining features such as the Mississippi River, the Appalachian Mountains, coastal plains, wetlands, and extensive agricultural lands, all of which have historically shaped the settlement patterns and economic foundations of the region.

Regarding infrastructure, these states benefit from an extensive network of highways and interstates, major port facilities (including Houston, New Orleans, and the Port of Virginia), as well as well-established rail and air transportation systems. Nonetheless, disparities in connectivity remain evident, particularly between densely populated urban centers and rural or mountainous areas. With respect to utilities such as electricity, telecommunications, and water, urban areas tend to enjoy high service quality and reliability, while rural and remote mountain communities continue to face notable deficiencies in service stability and accessibility.

In the socio-economic domain, the population reflects a rich demographic diversity. Younger cohorts are increasingly drawn to urban environments, whereas rural and mountainous areas are undergoing pronounced demographic aging. In terms of income and employment, states such as Virginia and North Carolina exhibit relatively high median incomes and educational attainment, whereas Mississippi and certain other sub-regions contend with persistently low-income levels and elevated poverty rates (United States Census Bureau, 2022). Economically, metropolitan areas tend to be anchored in industry, finance, and technology sectors—displaying

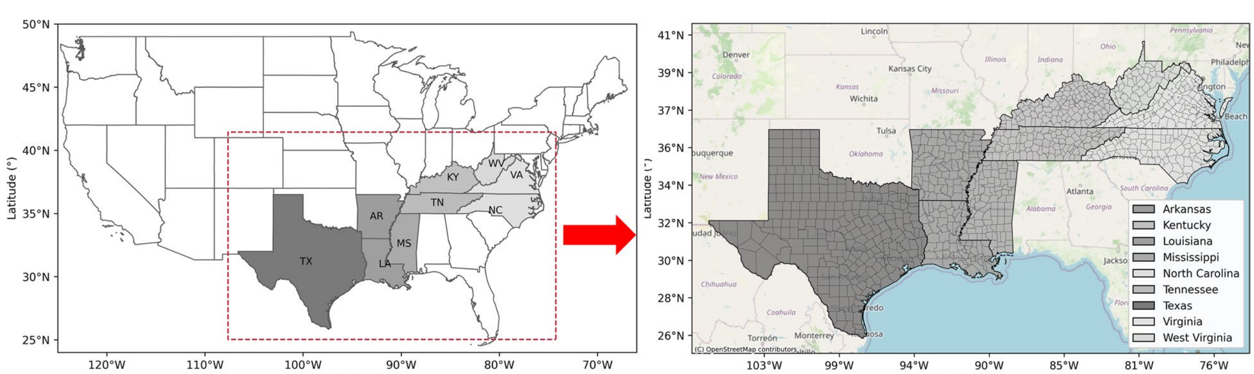

Fig. 1 Study area for state level (left) and county level (right)

trends of job growth and wage increases—while economic indicators for rural areas demonstrate a continued reliance on agriculture and low-wage industries, highlighting a persistent and structural urban–rural divide.

2.1 Power outages

Data related to power outages were compiled from a local news site (Caller-Times Corpus Christi, 2022). This power outage data is defined by the number of customers per county utility companies serve. Additionally, the data is recorded at 3-h intervals, occasionally at 2-h intervals, depending on the circumstances across nine states.

Power outage data includes more than 90% of county data for each state, except for TN and VA. For TN and VA, only about 70% of the county information is available (See Fig. 2). This may be because some outage data is missing, or it could be inferred that these states were less affected by Winter Storm Uri compared to others, resulting in less than 90% county coverage.

2.2 Weather features

Weather data were collected were collected from the official weather forecast website (Weather Underground, 2022) and incorporated into the database. Differences in observation scales between power outage data and weather data were noted during data collection. Power outages are recorded at the county level, whereas weather data are recorded at the city level. Therefore, data were collected based on whether the county seat (city) represents the entire county.

The weather data consists of both numerical and categorical variables. Numerical variables include Temperature (°F), Dew point (°F), Humidity (%), Wind speed (mph), Wind gust (mph), Pressure (inHg), and Precipitation (in). Additionally, categorical variables include wind direction and weather conditions.

Wind direction is recorded based on cardinal directions (e.g., N, NNW, NW, NNE, CALM, NE, ENE, W, E, S, SSE, WNW, ESE, WSW, SE, VAR-not constant, SSW, and SW.), while weather condition is broadly categorized into eight types (e.g., Fair, snow, cloudy, sleet, mix, freezing rain, rain, and storm). In consideration of the potential variability in outcomes across states, Texas (TX), Arkansas (AR), Louisiana (LA), Mississippi (MS), Tennessee (TN), Kentucky (KY), West Virginia (WV), Virginia (VA), and North Carolina (NC) were included for analysis.

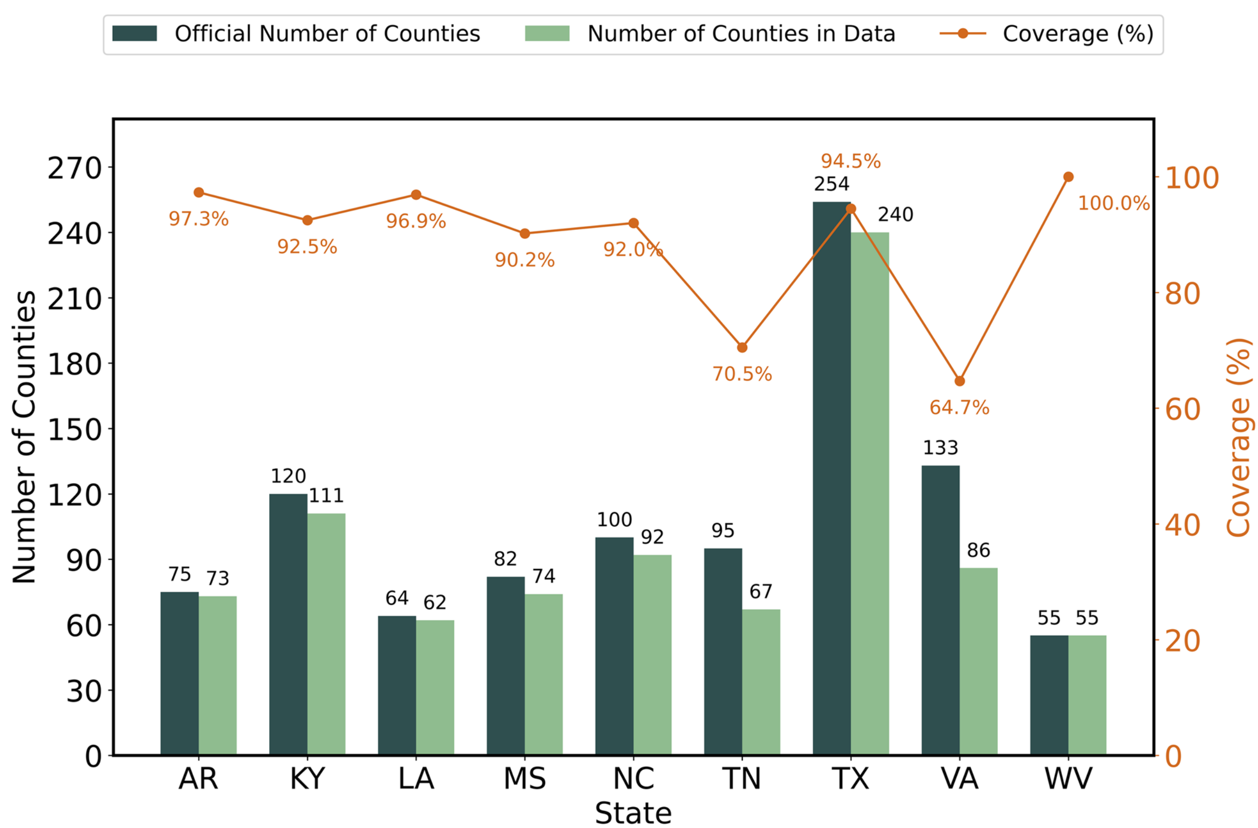

Fig. 2 Data coverage for the county level

2.3 Geographical dataset

Geographical data were obtained from the Texas Association of Counties (2022) and the United States Census Bureau (2022). These websites provided information about population, area, latitude, and longitude. The population was included in the dataset because it is considered a direct factor contributing to the over-demand of electricity, exacerbated by winter storm Uri-induced power outages. Therefore, population size was deemed to have a probable influence and was included as one of the variables.

Additionally, since the location of each county, indicated by latitude and longitude, might also have an impact, these variables were included. Furthermore, since area and population are closely related, area was also considered as an input variable. As the influence of geographical location may vary by state, it was also included as a categorical variable.

3 Data preprocessing and statistical descriptions

3.1 Target variable and predictor features

To create a predictive model for power outages caused by winter storm Uri, the target variable was "the number of power outages." More specifically, it is the number of customers experiencing power outages at the county level. However, in this study, it will be referred to as the number of power outages. These power outages were not recorded if the number of customers who experienced power outages was zero.

The independent variables consist primarily of meteorological data and geographical information, each further subdivided (See Table 1). Firstly, meteorological data encompasses Temperature, Dew Point, Humidity, Wind Speed, Wind Gust, Pressure, Precipitation, Wind Direction, and Weather Conditions. Wind Direction and Weather Conditions are categorical, while the remaining meteorological data are numerical. Geographical data includes Population, Area, Longitude, Latitude, and State. Population, Area, Longitude, and Latitude are numerical variables, while State is categorical. One-hot encoding was utilized to convert all categorical data into numerical vector data to effortlessly train the model.

Table 1 Summary of predictor variables for the present study

Category of predictor variable	Variables, *unit*	Types
Meteorological properties	Temperature,**°F**	Numerical
	Dew point,**°F**	
	Humidity,**%**	
	Wind speed, *mph*	
	Wind gust,*mph*	
	Pressure,*inHg.*	
	Precipitation,*in.*	
	Wind direction	Categorical
	Weather conditions	
Geographical properties	Population, *persons*	Numerical
	Area,mi^2	
	Longitude,°	
	Latitude,°	
	State	Categorical

The variables described above represent the foundational set considered in developing a preliminary outage prediction model for the southern United States. In contrast, in the northern region—particularly in the study by Cerrai et al. (2020)—the model was developed with a focus on Connecticut, a highly urbanized state in the Northeast U.S. In that region, power outages typically occur when heavy snowfall or strong winds from winter storms topple electric poles. Their findings identified the leaf area index, as an indicator of vegetation coverage, and snow accumulation as key predictive variables.

By comparison, in the South—and Texas in particular—such events occur infrequently and are not the primary cause of large-scale outages. Along the Texas Gulf Coast, hurricanes accompanied by flooding are more common and tend to have a significant impact on electrical infrastructure. In such cases, soil type has been shown to be an important factor in outage prediction, as indicated by the findings of Han et al., (2009a, 2009b).

Returning to the subject of winter storms, Texas experiences them rarely, and when major outages occur during such events, they often arise from an imbalance between electricity supply and surging demand rather than from direct physical damage to infrastructure. Moreover, the Electric Reliability Council of Texas (ERCOT), which operates the state's power grid, functions with limited interconnection to other states—a structural limitation that became a focal point of criticism during the 2021 winter storm event, when ERCOT was effectively isolated.

From a demographic perspective, Texas and Connecticut differ substantially. Texas contains 254 counties—the highest number in the United States—whereas Connecticut comprises only eight. According to United States Census Bureau (2022), the 2020 populations are 29,145,505 and 3,605,944, respectively, highlighting stark differences in geographic and demographic composition.

Returning once again to the matter of variable selection, the present study adopted only basic variables, given that the 2021 winter storm marked the first instance of a large-scale outage in Texas attributed to such an event. In future research, the inclusion of additional predictors such as leaf area index and soil type would enable a more nuanced assessment of their relative influence on outage occurrence.

3.2 Statistical information

Table 2 presents the basic statistical properties for the dependent and independent variables used in the predictive models. Each statistic represents the counties' mean and standard deviation within the respective state. There were an abundant number of data points from the state of Texas, likely due to the significant impact of winter storms.

Following Texas, the adjacent regions of Mississippi and Louisiana have 3,013 and 2,259 observed data points, respectively. Tennessee recorded the lowest average temperature, 11.64 °F, with a standard deviation of 24.26. Texas followed with the second lowest average temperature, 17.09 °F, and a standard deviation 20.31. Arkansas recorded the third lowest average temperature, 21.81 °F, with a standard deviation 11.63. For dew point, Arkansas had the lowest average, 13.37 °F, with a standard deviation of 11.63, followed by Texas and Kentucky. In terms of humidity, West Virginia had the highest average humidity, followed by Kentucky and Tennessee. Texas exhibited the highest average wind speed, followed by North Carolina and Louisiana. Similarly, Texas recorded the highest average wind gust, followed by North Carolina and Virginia.

West Virginia had the lowest average pressure, followed by Kentucky and Texas. North Carolina had the highest average precipitation, followed by Kentucky, Louisiana, Tennessee, and Virginia, all recording the same value of 0.01. Texas had the highest average county area, followed by Louisiana and Arkansas. Latitude and longitude information can be intuitively understood with the accompanying map of the study area. Population statistics represent average county populations. Tennessee had the highest average county population, followed by Texas and North Carolina.

All data is from February 11 to 23, 2021. The total number of observed data points is 27,832. In this context, data points represent the number of distinct observations rather than the magnitude of power outages.

For instance, consider a scenario where at time *t*, outages are recorded in three separate counties within a given state: County A reports five affected households, County B reports 3, and County C reports 1. In this case, we define this as comprising 3 data points, each corresponding to a unique county-level observation at the specified time *t*. It is crucial to note that the data points enumeration reflects the quantity of discrete

Table 2 Summary of statistical information for variables

State	AR	KY	LA	MS	NC	TN	TX	VA	WV
Data points, the number of observations	1,433	3,961	2,559	3,013	698	1172	***11,371***	1,609	2,026
Outages, the number of customer	150.24 (524.26)	1,159.67 (2,130.54)	1,418.68 (3,883.09)	1,202.65 (2,648.04)	138.65 (464.85)	249.37 (709.64)	***6,294.74*** (27,545.5)	513.27 (939.57)	1879 (4,451.48)
Temperature,**°F**	21.81 (14.19)	26.78 (9.86)	36.18 (13.85)	33.48 (14.31)	41.08 (9.83)	***11.64*** (24.46)	17.09 (20.31)	35.41 (8.91)	28.26 (7.85)
Dew point,**°F**	***13.37*** (11.63)	21.76 (8.58)	27.89 (10.29)	26.23 (10.79)	30.53 (8.3)	24.46 (11.26)	20.31 (13.62)	25.42 (8.76)	23.91 (7.73)
Humidity,**%**	72.21 (14.77)	82.74 (13.00)	75.35 (18.17)	78.3 (19.51)	70.85 (23.26)	76.98 (16.67)	72.74 (17.08)	70.80 (21.48)	***84.92*** (15.25)
Wind speed,***mph***	7.71 (4.45)	7.57 (4.15)	8.03 (4.88)	7.95 (4.34)	8.91 (4.56)	7.77 (4.37)	***11.14*** (5.55)	7.51 (5.16)	5.74 (4.23)
Wind gust,***mph***	1.33 (5.23)	2.04 (6.49)	2.41 (7.12)	2.17 (6.47)	3.62 (8.7)	2.05 (6.84)	***4.27*** (9.82)	3.29 (8.52)	2.81 (7.48)
Pressure,***inHg***	29.74 (0.41)	28.96 (2.38)	30.12 (0.16)	29.92 (0.17)	29.29 (0.74)	29.34 (0.38)	29.12 (1.34)	29.28 (8.52)	28.64 (7.48)
Precipitation,***in***.	0.00 (0.04)	0.00 (0.02)	0.01 (0.05)	0.01 (0.04)	***0.02*** (0.06)	0.01 (0.04)	0.00 (0.02)	0.01 (0.04)	0.00 (0.02)
Area,***mi***^{2}	744.72 (129.14)	342.87 (127.91)	779.72 (333.18)	617.66 (151.77)	551.82 (232.77)	490.97 (162.02)	***1,036.91*** (583.96)	461.87 (170.61)	469.89 (190.52)
Latitude,°	34.87 (0.87)	37.71 (0.50)	31.28 (1.03)	32.52 (1.13)	35.62 (0.55)	35.82 (0.46)	31.04 (1.80)	37.28 (0.51)	38.48 (0.68)
Longitude,°	−92.38 (1.15)	−83.93 (1.08)	−91.99 (1.10)	−89.82 (0.81)	−79.96 (1.83)	−85.86 (1.90)	−98.07 (2.34)	−79.01 (1.51)	−81.28 (0.86)
Population, 10^4, persons (new)	7.49 (10.62)	4.02 (9.20)	8.57 (11.08)	4.56 (5.45)	18.83 (25.80)	***21.38*** (28.49)	19.74 (55.55)	7.24 (16.15)	4.01 (3.92)

geographical units (counties) reporting outages, not the cumulative number of households experiencing power loss.

The overall mean and standard deviation of the comprehensive dataset are described in the following. The mean (standard deviation) for the number of outages is 3185.05 (17,920.80) households, temperature(°F) is 30.22 (14.68), dew point(°F) is 22.49 (12.01), humidity(%) is 75.88 (17.75), wind speed(mph) is 9.02 (5.28), wind gust(mph) is 3.13 (8.31), pressure(inHg) is 29.30 (1.52), precipitation(in.) is 0.5e-2 (0.03), area(mi^2) is 744.63 (485.19), latitude(°) is 33.59 (3.21), longitude(°) is −91.02 (7.09), and adjusted population is 123,993.99 (375,763.99).

3.3 Categorical information

The categorical variables consist of state, wind direction, and weather conditions. In Fig. 3, the histograms for each categorical variable are shown.

Most of the data is from the state of Texas, followed by Kentucky and Mississippi. North Carolina is the least represented in this dataset, indicating that the least

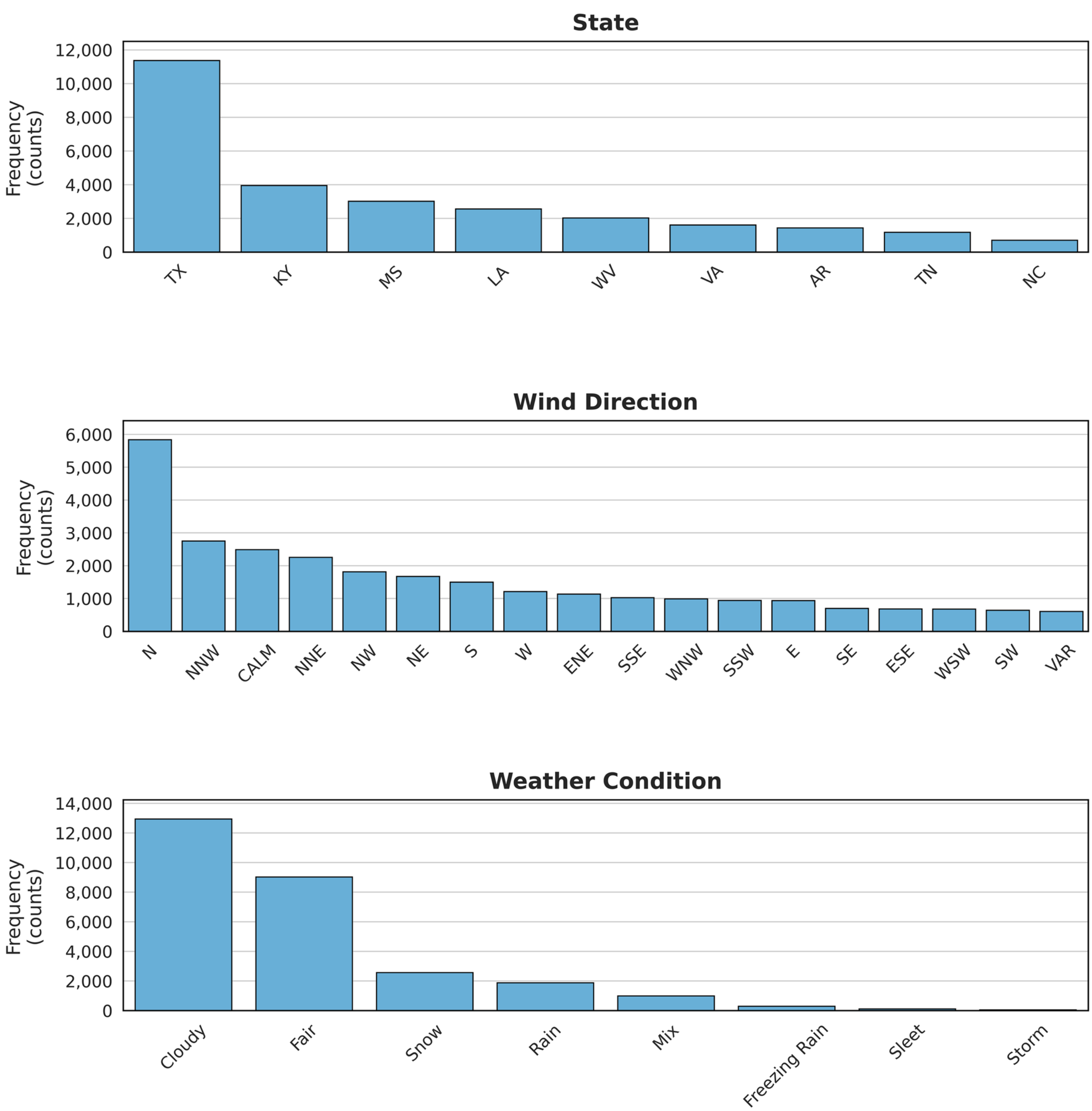

Fig. 3 Histogram of categorical variables

number of outages occurred in this state. Most of the wind was in the North (N) direction, followed by North–North-West (NNW) and the CALM category. The least data is for VAR, which indicates that the wind direction was difficult to define accurately at the time of measurement. Most of the data over the period studied was classified as the Cloudy weather condition, followed by fair and snow.

4 Methodology

Figure 4 shows the framework for this research. The purpose of this study is to develop a power outage prediction model for the southern region of the United States which has been greatly affected by Winter Storm Uri. Moreover, with this predictive model, the goal is then to utilize it to ascertain influential variables on outages. To achieve this, the response variable was defined as the number of power outages in each county. Initially, data on power outages, weather, and geographical factors were collected. Subsequently, an update on population information was required, wherein instances where the percentage of power outages exceeded the population were replaced by the maximum number of power outages.

Additionally, categorical variables were converted into numerical variables via one hot encoding. The dataset was split into training and testing sets, and six different models were trained. During the data splitting process, this study used 80% of the data as the training set and reserved the remaining 20% for the test set. This is because using 80% of the data for training achieves a balance between model learning and evaluation and is also a statistically valid choice for obtaining reliable performance estimates (Gholamy et al., 2018). Evaluation metrics were employed to compare the performance of these models. Following this, feature importance and partial dependence plots were utilized to examine which variables had the most significant impact on the models. Finally, a final model proposal was presented.

4.1 Nonparametric regression models

In this research, four non-parametric machine learning-based regression models are employed. The following subsection will briefly introduce the models: Random Forest, XGBoost, LightGBM, and CatBoost. These models can be implemented for classification or regression; however, as this work focuses on predicting the actual number of outages in each county (a regression task), the following subsections will discuss the regression version of each model.

4.1.1 Random Forest

The Random Forest model is an adequate tool for predicting power outages or damage from extreme weather

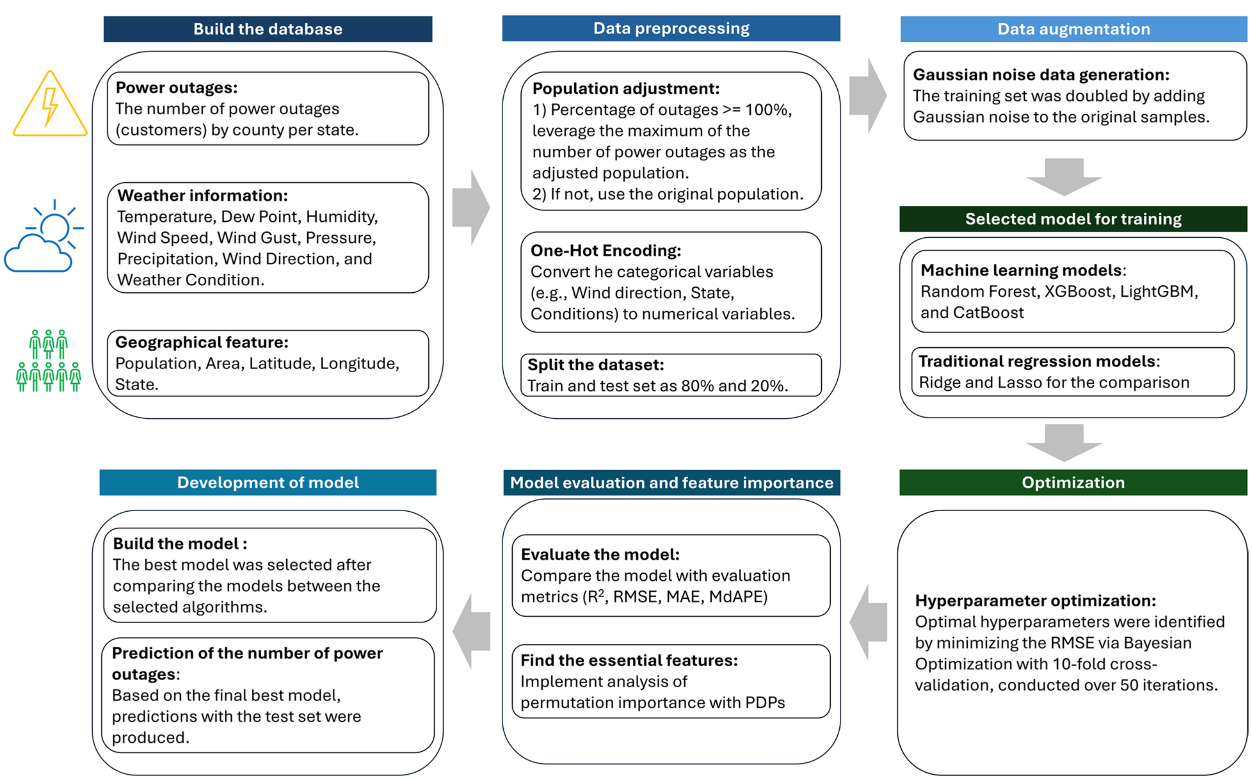

Fig. 4 Framework for this study

conditions due to its ability to handle complex, non-linear relationships between input and output variables and its robustness to missing and noisy data. It is an ensemble learning method that combines multiple decision trees to make more accurate predictions. Each decision tree is trained on a random subset of the input variables and data samples, and the output of the Random Forest model is the average or majority vote of the outputs of the individual trees. The key idea behind Random Forest is that combining multiple decision trees can reduce the variance and overfitting that can occur with a single decision tree. Random Forest also allows us to estimate the importance of each feature in making predictions. Feature importance scores can be calculated by measuring the decrease in accuracy when a feature is randomly permuted (Breiman, 2001; James et al., 2013).

4.1.2 XGBoost

XGBoost (Extreme Gradient Boosting), is a powerful machine learning algorithm that has been widely used in predictive modeling, particularly in the field of natural disaster management. Like Random Forest, XGBoost is also an ensemble learning algorithm. It combines multiple weak models to create a strong model with high accuracy (Chen & Guestrin, 2016). Moreover, XGBoost also uses decision trees as the base model, which can capture both linear and nonlinear relationships between the predictors and the outcome variables (Chen & Guestrin, 2016).

Additionally, XGBoost implements gradient boosting, an iterative process that sequentially adds new weak models to the existing model to improve its performance (Chen & Guestrin, 2016). XGBoost is a flexible and highly scalable method (Daoud, 2019). According to Daoud (2019), XGBoost differs from other gradient-boosting techniques because a new regularization method is employed to address overfitting problems. Therefore, model tuning can be implemented to speed up the training, and thus, XGBoost shows more robust performance than comparable methods often. For this reason, the XGBoost model will be considered as a candidate for predicting power outages during winter storms in this work. The objective function is expressed in (1):

$$f_o = \sum_{i=1}^{n} \mathcal{L}\left(y_i, f(x_i)\right) + \sum_{j=1}^{m} \mathcal{R}\left(f_j\right) + C \tag{1}$$

where y_i indicates the label or target value, $x_i = (x_{1i}, x_{2i}, \ldots, x_{ri})$ represents the indicators, $\mathcal{L}$ is a loss function, $f(x_i)$ explains the predicted value, $\mathcal{R}(f_j)$ represents the regularization at the jth iteration, and C is a constant value which can be zero.

The regularization function can be defined as (2):

$$R\left(f_j\right) = \gamma T + \frac{1}{2}\lambda \sum_{k=1}^{T} w_k^2 \tag{2}$$

where γ denotes the leaf complexity, T represents the number of leaves, λ is a parameter for the penalty, and w_k is a result of the output for each node of the leaf.

4.1.3 LightGBM

Ke et al. (2017) first introduced the LightGBM-based regression model, which reduces the implementation time of the training process. This algorithm is also based on the decision tree, but it differs from the classic decision tree in that it does not check all of the last leaves for each new leaf. According to Daoud (2019), it is designed to handle large datasets with high-dimensional feature spaces and can achieve high accuracy with less memory usage and faster training speed compared to other gradient boosting algorithms such as XGBoost and CatBoost. For all of these reasons, it is a strong candidate for estimating power outages in this study.

The LightGBM algorithm uses histogram-based decision tree and leaf-wise tree methods with a maximum depth to enhance training speed and reduce memory requirements (Liang et al., 2020). Unlike traditional decision trees, LightGBM uses a level-wise growth strategy where leaves on the same layer are split simultaneously. The leaf-wise tree method selects the leaf with the maximum information gain at each step, rather than splitting the tree level-by-level. It is worth noting that leaves on the same layer may not have the same information gain, which is the estimated reduction in entropy that occurs by dividing the nodes (Kodaz et al., 2009). Therefore, LightGBM uses information gain to determine the most efficient way to split the leaves and achieve improved model accuracy. This information gain can be expressed as (3), and the entropy information for the collection C can be calculated by (4).

$$I(C, A) = E(C) - \sum_{v \in A} \frac{|C_v|}{C} E(C_v) \tag{3}$$

$$E(C) = \sum_{n=1}^{N} -p_n \log_2 p_n \tag{4}$$

where $E(C)$ represents the entropy information for the collection C, p_n denotes the portion of C related to category n, N indicates the number of categories, v is the attribute value A, and C_v represents the subset for C, where the attribute has a value v.

4.1.4 CatBoost

Dorogush et al. (2018) introduced CatBoost, a novel gradient boosting algorithm for regression that is particularly well-suited for handling categorical features. Traditional gradient boosting algorithms typically convert categorical features to numerical values through one-hot encoding, which can result in a large number of features and computational inefficiency. In contrast, CatBoost uses an efficient algorithm for handling categorical features, converting them to numerical values using an ordered boosting algorithm. This approach results in more compact models and improved generalization performance, especially on datasets with high-cardinality categorical features. The CatBoost algorithm is known for addressing several combinations of features and is advantageous in preventing overfitting. Given these advantages and its ability to handle categorical features efficiently, the CatBoost model is a strong candidate for predicting power outages triggered by winter storms.

4.2 Traditional statistical models

In this study, in addition to the four non-parametric machine learning models, Ridge regression and Lasso regression models were also included for comparison. The concepts of simple/multiple linear regression, Ridge regression, and Lasso regression are well documented in the literature (James et al., 2013; Tibshirani, 1996).

Simple Linear Regression is a statistical method utilized to examine how a single independent variable influences a dependent variable (Roustaei, 2024). It identifies the straight line that best represents the relationship between the two variables within the data. This line can then be used to predict the value of one variable given the value of the other.

Multiple Linear Regression is a statistical method used to analyze the simultaneous effects of various independent variables on a dependent variable (Jobson, 1991). This approach models the relationship between several predictors and the outcome, enabling the construction of a predictive model in which each independent variable may have a distinct influence on the dependent variable.

Ridge Regression is a statistical model that employs a technique to address issues arising from a high degree of multicollinearity among independent variables or excessive model complexity leading to overfitting. Multicollinearity refers to the phenomenon in which two or more independent variables in a regression analysis exhibit a strong linear correlation with each other. By introducing a regularization term that penalizes significant regression coefficients, Ridge Regression constrains the magnitude of these coefficients, thereby encouraging a more parsimonious model structure (Hoerl & Kennard, 1970). This regularization facilitates more stable and reliable parameter estimation and, consequently, enhances the predictive performance of the model.

Lasso Regression is a statistical model that uses a regularization technique designed to prevent overfitting while simultaneously enabling automatic variable selection by shrinking the coefficients of less significant or irrelevant predictors to precisely zero (Tibshirani, 1996). Like Ridge Regression, Lasso imposes a penalty on the magnitude of the regression coefficients; however, its unique ability to reduce some coefficients to zero facilitates the identification and retention of only the most significant variables. This property results in a sparser and more interpretable model, enhancing both model simplicity and interpretability.

4.3 Data augmentation

In this study, the development of the model was hindered by limited accessibility to data beyond the singular winter storm event. As an alternative, a data augmentation method involving the injection of 1% Gaussian noise was adopted. Gaussian noise is a probabilistic random noise that follows a normal (Gaussian) distribution. In this approach, for a training dataset containing N samples, an additional N samples were generated, resulting in a total of 2N training data points. In each augmented sample, Gaussian noise with a standard deviation of 1% was independently added to each feature. This augmentation process was conducted exclusively during the training phase, prior to model evaluation.

During the K-Fold Cross-Validation procedure (with K = 10 in this study), 20% of the original dataset was randomly selected as the test set and left unaltered. The remaining 80% was used for training and validation. For each fold, one of the ten subsets was randomly selected as the validation set, and the remaining nine subsets were augmented using the above method. This ensured that no augmented data was included in the validation process, thereby maintaining the objectivity and fairness of model assessment.

Although this noise-based augmentation method contributed to enhancing the model's credibility, it is anticipated that the incorporation of additional historical outage events or scenario-based synthetic outage data could further improve both the external validity and credibility of future models.

4.4 Evaluation metrics

4.4.1 Coefficient of determination, R^2

The coefficient of determination is known as the R^2 score and is widely used in regression methods at the evaluation stage. By using this evaluation metric, researchers can intuitively know the accuracy of the regression models. This value can be calculated by using the following equation:

$$R^2 = 1 - \left[\frac{\Sigma \left(y - \hat{y} \right)^2}{\Sigma \left(y - \bar{y} \right)^2} \right] \tag{5}$$

where y is the sample value, $\hat{y}$ is the predicted value, and $\bar{y}$ is the mean of the samples.

4.4.2 Root mean squared error, RMSE

Root mean squared error (RMSE) is also widely used to evaluate the performance of machine learning models, and this value is more commonly used than the mean squared error (MSE) because RMSE can be similar to the error between predicted and actual values. Furthermore, the RMSE value can be obtained by the following equation:

$$RMSE = \sqrt{\left(\frac{1}{n} \right) \Sigma \left(\mathrm{y} - \hat{y} \right)^2}$$

where n is the number of samples, y is the sample value, and $\hat{y}$ is the predicted value.

4.4.3 Mean absolute error, MAE

The mean absolute error is similar to the mean squared error because of scale. However, the MAE value differs from RMSE in that this metric uses the sum of the absolute value of the difference between predicted and actual values instead of the sum of the squared difference. The MAE value can be calculated via the following equation:

$$MAE = \left(\frac{1}{n} \right) \Sigma \left| y - \hat{y} \right| \tag{6}$$

where n is the number of samples, y is the sample value, and $\hat{y}$ is the predicted value.

4.4.4 Median absolute percentage error, MdAPE

As expressed in (8) below, the median absolute percentage error (MdAPE) calculates the median of the absolute percentage of the errors. This metric is less sensitive to outliers than the Mean Absolute Percentage Error, rendering it particularly advantageous when outlier data points may skew the evaluation.

$$MdAPE = median\left(\left| \frac{y_1 - \hat{y}_1}{y_1} \right|, \left| \frac{y_2 - \hat{y}_2}{y_2} \right|, \ldots, \left| \frac{y_N - \hat{y}_N}{y_N} \right| \right) \tag{7}$$

where N is the number of total samples, y_N is the sample value for Nth, and $\hat{y}_N$ is the predicted value for Nth.

4.5 Data split

Determining the ratio between the training set and the test set is a simple yet challenging task in model development. Gholamy et al. (2018) conducted a study using mathematical and statistical methods to demonstrate which ratio is the most optimal for splitting the data into training and test sets. To prove this, they used the function $f(p)$ which describes how the variance of the final prediction error (approximation error) changes with respect to the sample size N when the training and test data are split in a ratio $p : (1 - p)$.

$$f(p) = \frac{1}{pN} + \frac{1}{(1-p)N} = \frac{1}{p(1-p)N} \tag{8}$$

where p is the proportion of the training set within the entire dataset, and N is the total number of data samples (i.e., the total dataset size).

Their research argued that the total error of the model on the test data becomes minimized when the proportions of training and testing data are properly balanced. According to the theoretical analysis, the optimal value is 50%. However, in practice, if the training data is insufficient, the performance of machine learning models may degrade. Therefore, they set a condition that the confidence interval (uncertainty) of the test error should be at least twice that of the training error for the evaluation to be considered reliable. This condition can be expressed as a formula below using the proportion of training data and the total number of data points. In the equation, oneleft) is the approximate magnitude of the error (uncertainty) estimated from the training dataset, or its confidence interval. (This value decreases as the number of samples increases, i.e., as pN becomes larger) and $1/\sqrt{(1-p)N}$ (right) is the magnitude of the error (uncertainty) estimated from the test data. As a result, they concluded that 0.8 (i.e., 80% training and 20% testing) is the most reasonable ratio.

$$2 \bullet \frac{1}{\sqrt{pN}} \leq \frac{1}{\sqrt{(1-p)N}} \tag{9}$$

Therefore, their splitting method is a statistically valid choice for obtaining reliable performance estimates, so the training/test ratio was set to 80:20.

4.6 Cross-validation

Utilizing K-fold cross-validation enables a robust evaluation of the model's ability to generalize (Raschka et al., 2022). This paper used a tenfold cross-validation procedure to demonstrate the generalization capabilities of the resulting models. This study focused on implementing cross-validation using the coefficient of determination. As shown in Fig. 5, the entire training dataset is arbitrarily divided into ten subsets of the same size. One of the ten subsets is employed as the validation set, and the training subsets consist of the other nine subsets. This procedure is repeated ten times with a subset selected as the validation set in each run.

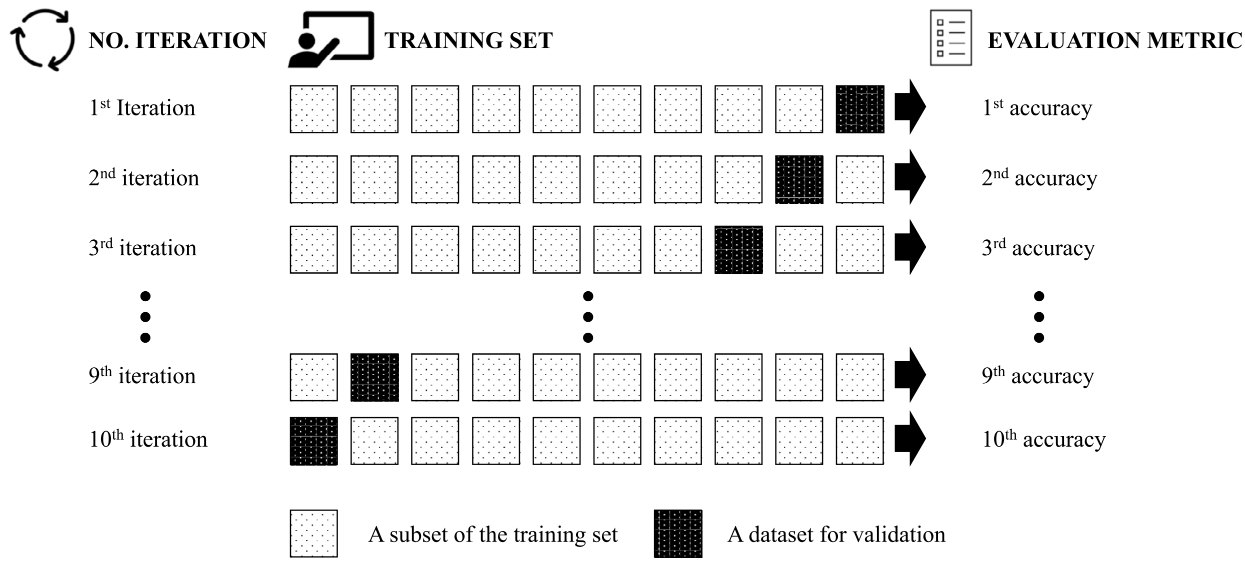

Fig. 5 A scheme for 10 folds cross-validation

Although it may seem straightforward to simply split the training data by a fixed ratio into separate training and validation sets, determining the optimal allocation is actually quite challenging. For this reason, Kohavi (1995) evaluated model performance using various values of k (such as 2, 5, 10, 20, ...)-fold cross-validation. Their results showed that while 2- or fivefold cross-validation is computationally faster, it introduces larger bias and can reduce overall accuracy. In contrast, tenfold cross-validation achieves high accuracy with low variance and, when compared to folds greater than 20, is the most efficient choice. With higher k values in a k-fold cross-validation, it does not significantly improve model performance. Therefore, in this study, we employed tenfold cross-validation for model evaluation.

4.7 Bayesian optimization

This work applies the Bayesian optimization technique to optimize the hyperparameters of the four machine learning models in efforts towards selecting the best model. Bayesian optimization is a common technique used to maximize the performance of machine learning models (Wang et al., 2021). Bayesian optimization has the advantage of being able to optimize with fewer attempts than more exhaustive methods such as the Grid Search algorithm. Therefore, the Bayesian optimization technique was adopted for all four models in this work considering its quick capabilities as speed is of utmost importance for power failure prediction in the face of natural hazards. The key objective of Bayesian optimization is to maximize or minimize the objective function (R2 in this case). The parameters below represent the range and hyperparameters for each model (See Table 3). In addition, ten-fold cross-validation was conducted, and the search process was repeated 50 times to find the optimal hyperparameters for each of the four models.

4.8 Partial dependence plot, PDP

Partial dependence plots (PDPs) were first proposed by Friedman (2001). Friedman (2001) developed PDP to improve the interpretability of machine learning models, especially complex ensemble models.

A PDP is a graph that visualizes the average effect of a specific input feature on the predicted outcome of a machine learning model. PDP is particularly useful for interpreting models that are difficult to understand or are considered "black-box", such as ensemble models, by helping to reveal the internal decision-making process.

The algorithm can be described as follows. First, PDP changes the value of one or two features of interest, while keeping all other features constant. Second, for each data point, it varies the value of the selected feature and calculates the model's prediction, then averages these predictions. Finally, the plot shows how the predicted value changes as the feature value changes, illustrating the feature's marginal effect. Marginal effect means "the average effect of a single feature on the prediction, after averaging out the influence of all other features." More detailed information can be available from the original paper from Friedman (2001).

Table 3 Parameters selected based on Bayesian optimization

Model, parameter, and range/options	
Random forest - *'n_estimators'*: (100, 500) - *'max_depth'*: (3, 15) - *'min_samples_split'*: (2,10) - *'min_samples_leaf'*: (1,10) - *'max_features'*: (None, 'sqrt', 'log2') - *'bootstrap'*: (True, False)	**XGBoost** - 'n_estimators': (100, 500) - 'max_depth': (3, 15) - 'learning_rate': (0.01, 0.3, 'log-uniform') - 'subsample': (0.5, 10) - 'colsample_bytree': (0.5, 10) - *'min_child_weight'*: (1, 10) - *'gamma'*: (0, 5) - *'reg_alpha'*: (0, 5) - *'reg_lamda'*: (0, 5)
LightGBM - *'n_estimators'*: (100, 500) - *'max_depth'*: (3, 15) - *'learning_rate'*: (0.01, 0.3, 'log-uniform') - *'num_leaves'*: (20, 100) - *'min_child_samples'*: (5, 30) - *'subsample'*: (0.5, 1.0) - *'colsample_bytree'*: (0.5, 1.0)	**CatBoost** - *'literations'*: (100, 500) - *'depth'*: (3, 10) - *'learning_rate'*: (0.01, 0.3, 'log-uniform') - *'l2_leaf_reg'*: (1, 10) - *'border_count'*: (32, 255) - *'bagging_temperture'*: (0, 1)
Ridge - *'alpha'*: (10^{-4}, 10)	**Lasso** - *'alpha'*: (10^{-3}, 10^{3})
Description of parameter	
Random forest - *'n_estimators'*: The number of trees in the forest - *'max_depth'*: The maximum depth of each tree - *'min_samples_split'*: The minimum number of samples required to split an internal node - *'min_samples_leaf'*: The minimum number of samples required to be at a leaf node - *'max_features'*: The number of features to consider when looking for the best split - *'bootstrap'*: When bootstrap samples are used when building trees.'True' means samples are drawn with replacement	**XGBoost** - 'n_estimators': The number of gradient-boosted trees to be built - 'max_depth': Maximum depth of a tree - 'learning_rate': The step size shrinkage used in updates to prevent overfitting - 'subsample': The fraction of samples to be used for each tree - 'colsample_bytree': The fraction of features to be used fro each tree - *'min_child_weight'*: Minimum sum of instance weight (hessian) needed in a child - *'gamma'*: Minimum loss reduction required to make a further partition on a leaf node - *'reg_alpha'*: L1 regularization term on weights - *'reg_lamda'*: L2 regularization term on weights
LightGBM - *'n_estimators'*: The number of boosting iterations - *'max_depth'*: Maximum depth of a tree - *'learning_rate'*: Controls the contribution of each tree to the final prediction - *'num_leaves'*: The maximum number of leaves in one tree - *'min_child_samples'*: Minimum number of data needed in a child (leaf) - *'subsample'*: Fraction of data to be used for each iteration - *'colsample_bytree'*: Fraction of features to be used for each iteration	**CatBoost** - *'literations'*: The number of boosting iterations - *'depth'*: Depth of the tree - *'learning_rate'*: The step size shrinkage used in updates to prevent overfitting - *'l2_leaf_reg'*: L2 regularization term on weights - *'border_count'*: Number of splits for numerical features - *'bagging_temperture'*: Controls the intensity of Bayesian bagging
Ridge - *'alpha'*: In linear regression, it is a hyperparameter used for regularization that controls the strength of the constraints placed on the coefficients	**Lasso** - *'alpha'*: In linear regression, it is a hyperparameter used for regularization that controls the strength of the constraints placed on the coefficients

Notes: L1 regularization simplifies the model and removes unnecessary variables (feature selection effect). L2 regularization constrains all model parameters to reduce overfitting (coefficients are not reduced completely to zero)

5 Results and discussion

This section displays the results from the optimized training and testing of the four machine learning models and the Ridge and Lasso regression models. Further, a permutation importance procedure was carried out and the results are presented.

5.1 Optimized model and cross-validation

Figure 6 presents the cross-validation results for six different models, including the R^2, RMSE, MAE, and MdAPE metrics. In addition, the 95% confidence intervals were calculated based on the mean and standard deviation obtained from the 10 folds. The results demonstrate that, in terms of the R^2 metric, XGBoost, CatBoost, and LightGBM achieved values close to 1.00, suggesting exceptionally strong model fitting during the training process. It is presumed that these results indicate a high degree of learning within the training phase. Consequently, a subsequent section will verify these findings using an independent test set, with further details to be provided in that section.

From the RMSE results in Fig. 6, XGBoost recorded the lowest value at 849.66 households, indicating the best

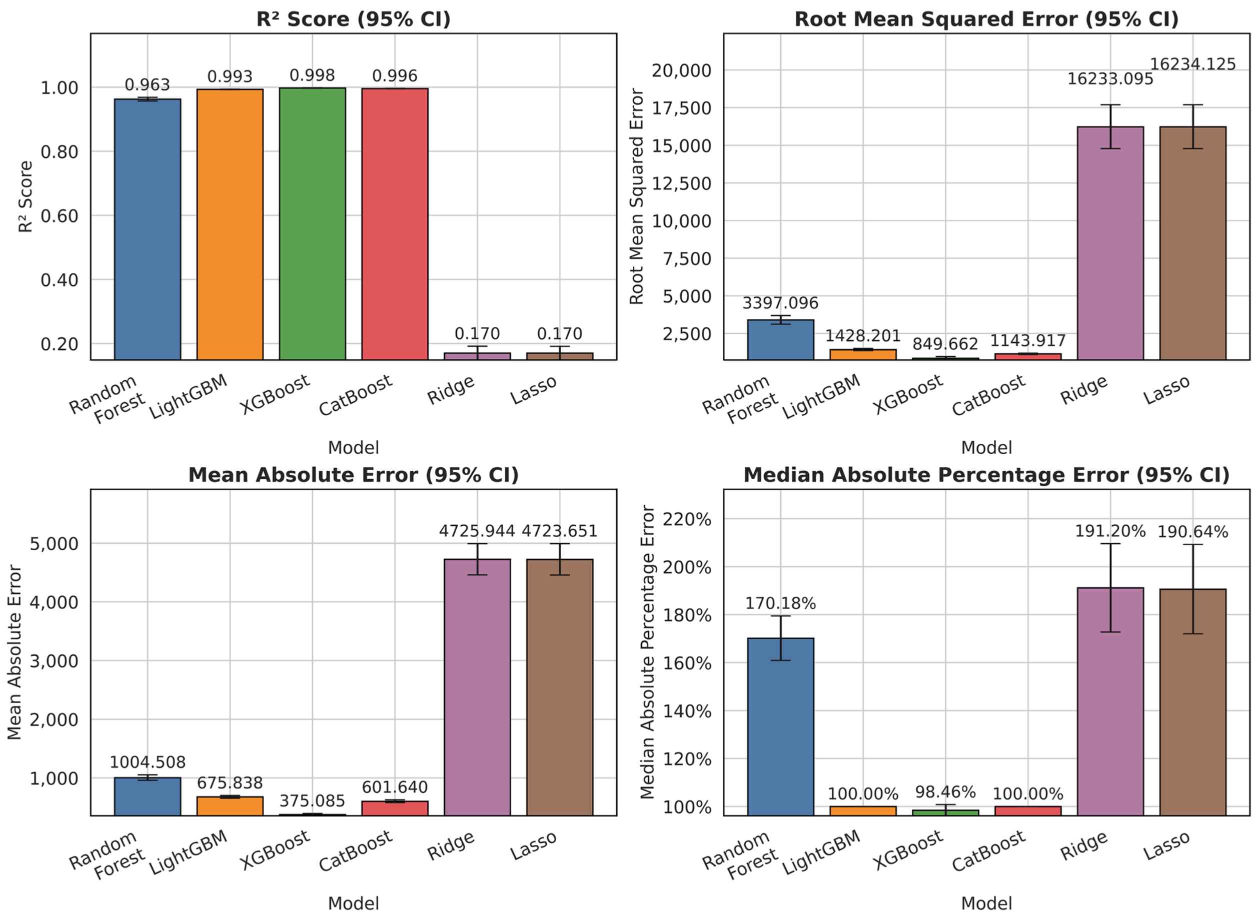

Fig. 6 Cross-validation results during optimization

performance among the evaluated models. By contrast, traditional statistical models such as Ridge and Lasso displayed considerably higher RMSE values of 16,233.10 and 16,234.13 households, respectively, which are substantially larger compared to those of the four more advanced machine learning models. This suggests that the machine learning approaches—particularly XGBoost, LightGBM, and CatBoost—outperformed the traditional regression techniques in terms of predictive accuracy.

In the MAE and MdAPE results presented in Fig. 6, XGBoost recorded values of 375.09 households and 98.46%, respectively. These outcomes align closely with the previously discussed R^2 and RMSE findings, further reinforcing the conclusion that XGBoost consistently delivered superior performance among all models examined in this study.

The optimal parameters for each model are detailed below, and these parameter configurations were subsequently employed in the later section to evaluate each model's performance on the test set.

Random Forest: 'n_estimators': 243, 'max_depth': 15, 'min_samples_split': 2, 'min_samples_leaf': 1, 'max_features': 'none, 'bootstrap': False

XGBoost: 'n_estimators': 314, 'max_depth': 14, 'learning_rate': 0.08, 'subsample': 0.85, 'colsample_bytree': 0.75, *'min_child_weight'*: 9.60, *'gamma'*: 1.88, *'reg_alpha'*: 5.00, *'reg_lamda'*: 1.70

LightGBM: *'n_estimators'*: 248, *'max_depth'*: 11, *'learning_rate'*: 0.3, *'num_leaves'*: 46, *'min_child_samples'*: 10, *'subsample'*: 1.00, *'colsample_bytree'*: 1.00

CatBoost: *'literations'*: 384, *'learning_rate'*: 0.30, *'l2_leaf_reg'*: 1.84, *'border_count'*: 109, *'bagging_temperture'*: 1.00

Ridge regression: *'alpha'* : 7.85

Lasso regression: 'alpha' : 1.67

5.2 Model performance

The purpose of this study is to develop a power outage prediction model for the nine states damaged by winter storm Uri. In this study, 80% of all data was set as the training set, with the remaining 20% held out as the test set. Each optimized model was tested using the test set

(See Fig. 7), and the results are shown below using the four evaluation metrics introduced previously (Table 4). In addition, if the prediction result is less than 0, "the number of power outages" is set to 0 because it cannot be less than 0. Consistent with the cross-validation (CV) results, XGBoost achieved the highest R^2 value of 0.92, along with the lowest RMSE (4,991.9 households) and MAE (1,192.1 households), thereby demonstrating the best overall predictive performance among the models. However, in the case of MdAPE, XGBoost exhibited a value of 100%, identical to that of CatBoost and LightGBM.

Ridge and Lasso regression models displayed results similar to those observed in the CV analysis, confirming

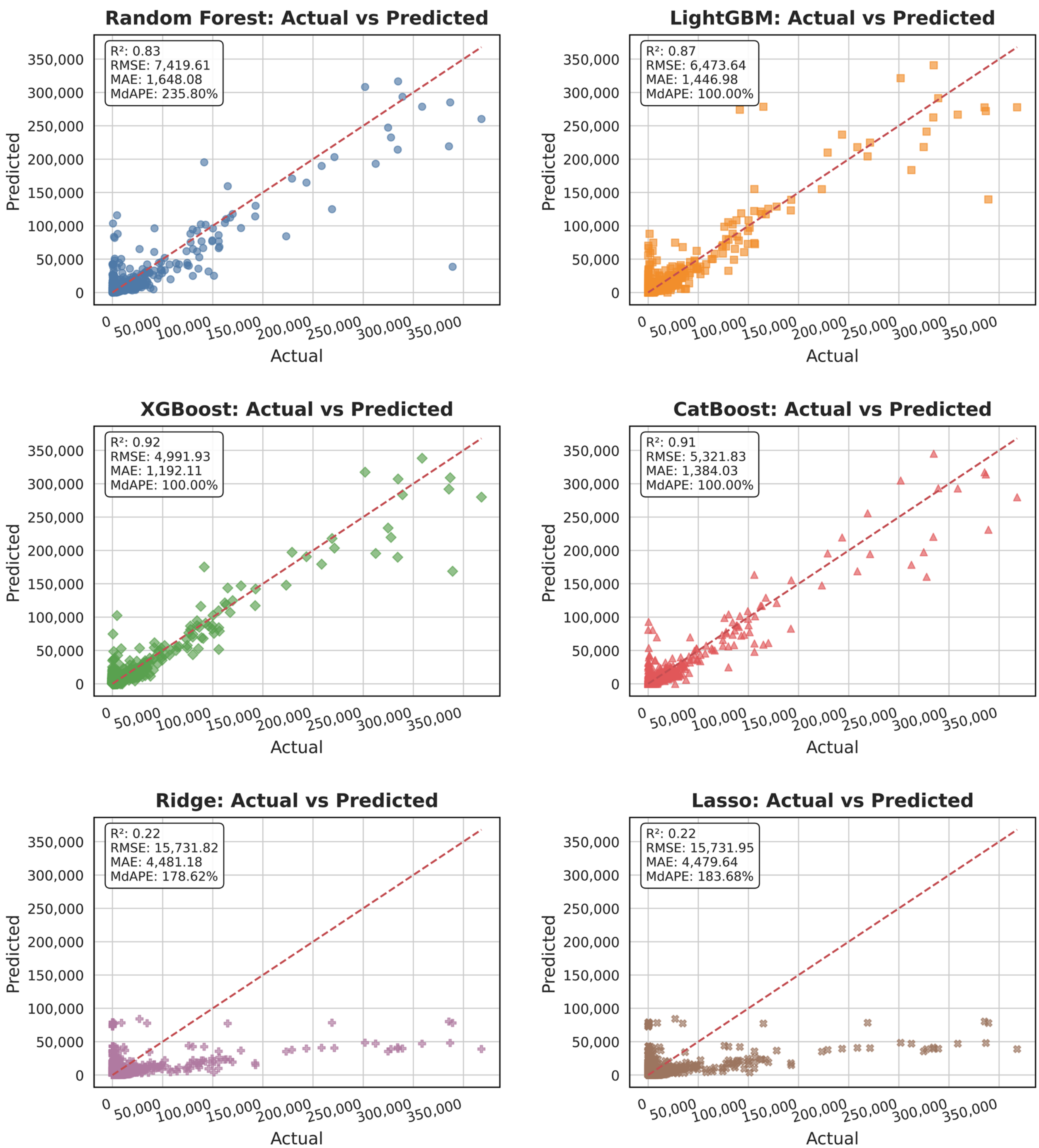

Fig. 7 Actual vs. Predicted values with optimized models

Table 4 Evaluation results for the testing set

Model	R^2	RMSE	MAE	MdAPE (%)
Random Forest	0.83	7,419.6	1,648.1	235.8
XGBoost	0.92	4,991.9	1,192.1	100.0
LightGBM	0.87	6,473.6	1,446.9	100.0
CatBoost	0.91	5,321.8	1,384.0	100.0
Ridge	0.22	15,731.8	4,481.2	178.6
Lasso	0.22	15,731.9	4,479.6	183.7

their comparatively weaker performance. Specifically, both models achieved an R^2 of 0.22, with RMSE values of 15,731.8 and 15,731.9 households, and MAE values of 4,481.18 and 4,479.64 households, respectively. Nonetheless, for MdAPE, their results—178.6% for Ridge and 183.7% for Lasso—were lower than that of Random Forest, which recorded a substantially higher accuracy compared to MdAPE of 235.8%.

5.3 Weighted evaluation metrics

In this study, the test dataset contained varying numbers of samples across different states. Therefore, to account for this imbalance in the dataset, a weighted evaluation metric was implemented. The test set, encompassing all states, consisted of 5,567 data points. Accordingly, weights were calculated based on the number of samples in each state, as presented in the Table 5 and these weights were applied to compute the weighted evaluation metric.

Examining the data in Table 5, it is observed that, within the test set, the state of Texas (TX) contains the largest number of data points, totaling 2,270 instances. This is followed by Kentucky (KY), with 799 instances. Based on the total of 5,567 samples in the test set, the proportion of each state's data was calculated and used to derive corresponding weights. As a result, TX was assigned a weight of 0.41, KY a weight of 0.14, and the remaining states are summarized in Table 5.

Table 5 State-wise data counts and weights

State	Data Count	Weight
AR	266	0.05
KY	799	0.14
LA	520	0.09
MS	586	0.11
NC	145	0.03
TN	236	0.04
TX	2,270	0.41
VA	356	0.06
WV	389	0.07

Using these calculated weights, the test results were recomputed, as presented in Table 6. A comparison with the results of Table 4—which were computed without weighting—reveals that all performance metrics decreased when weights were applied. Notably, even XGBoost, which demonstrated the highest performance in Table 4, showed a decline, registering an R^2 of 0.74 along with increases in other error metrics, indicating a reduction in predictive accuracy. This decline in performance is likely attributable to the increased weight assigned to extreme outage cases. For instance, when predicting such extreme cases within Texas, larger prediction errors appear to have been amplified by the higher weight, ultimately leading to a further decrease in overall accuracy.

5.4 Permutation importance

The permutation importance was applied to examine which variables are the most influential as predictors of power outages during a winter storm. Figure 8 shows the feature importance results of each model. This graph includes the results of each subcategory for categorical variables (e.g., wind direction, weather conditions, and state).

The permutation importance for each feature within each model was computed using the RMSE metric as the basis for evaluation. For every feature, the calculation was repeated ten times, and the resulting values were averaged. These average values were then normalized via the min–max standardization method to derive the final permutation importance scores for each model, as illustrated in Fig. 8. For categorical features, the permutation importance was computed by comparing model performance when the variable was present versus when it was absent.

In the min–max normalization process, the minimum value across the dataset was subtracted from each individual data point, and the result was divided by the range, defined as the difference between the maximum and minimum values. This transformation scales the smallest value to 0, the largest to 1, and proportionally maps all other values between 0 and 1.

The result of performing permutation importance ten times was derived. The bar graph represents the decrease in average accuracy, and the black scale represents the standard deviation range (See Fig. 8). Across the four machine learning models—XGBoost, LightGBM, CatBoost, and Random Forest—population, dew point, and pressure were identified as the most influential features. In contrast, for Ridge and Lasso, although their predictive accuracy was substantially lower, latitude and longitude, in addition to population, emerged as the most impactful variables.

Across the four machine learning models—XGBoost, LightGBM, CatBoost, and Random Forest—population,

Table 6 Results of the weighted evaluation metrics for each model

Model	Weighted R^2	Weighted RMSE	Weighted MAE	Weighted MdAPE (%)
Random Forest	0.67	5,445.4	1,648.1	279.7
XGBoost	0.74	3,782.3	1,190.1	101.9
LightGBM	0.59	4,814.1	1,447.0	110.8
CatBoost	0.53	4,061.3	1,384.0	100.0
Ridge	−4.05	11,899.4	4,481.2	359.6
Lasso	−4.07	11,898.4	4,479.6	369.6

dew point, and pressure were identified as the most influential features. In contrast, for Ridge and Lasso, although their predictive accuracy was substantially lower, latitude and longitude, in addition to population, emerged as the most impactful variables. These results partially explain the massive power outages caused by the failure of electricity production facilities using wind energy and natural gas to perform their roles due to the extreme cold weather in the face of Winter Storm Uri (Popik & Humphreys, 2021). This insufficient energy production failed

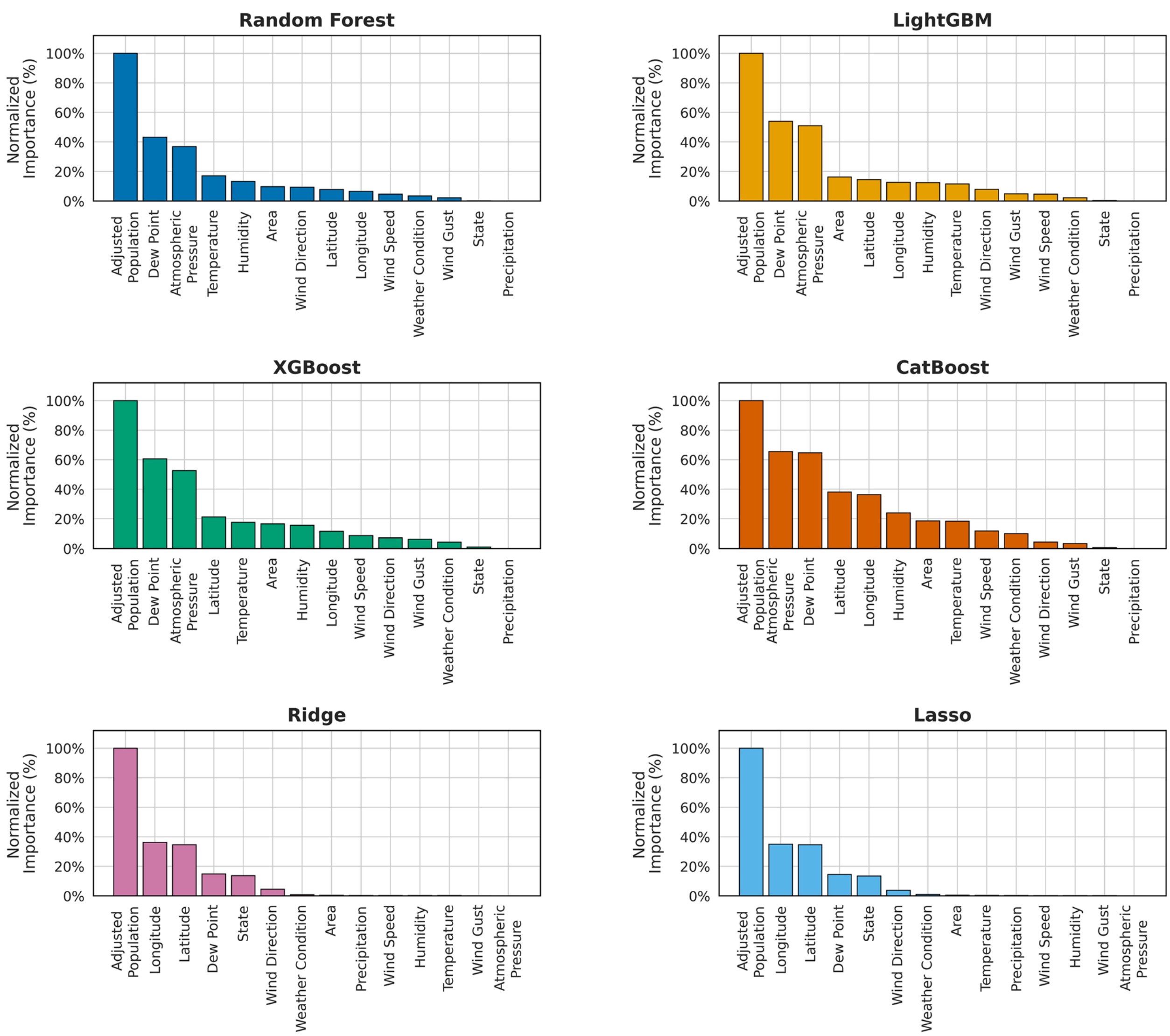

Fig. 8 Feature importance for each model

to meet the significantly increased demand, ultimately leading to large-scale power outages.

In addition, a more detailed interpretation of the results from the highest-performing XGBoost model, as presented in Fig. 8, is provided. The severe winter storm of 2021 was generated by a strong low-pressure system accompanied by rising air currents. During such winter storms, the dew point generally tends to decrease. Since the dew point is determined by the amount of water vapor in the atmosphere, it becomes lower when the air is colder and drier. In other words, during winter storms, reduced atmospheric moisture content results in a lower dew point, subsequently influencing the formation of snow or frost.

Within this context, the permutation results indicating population, dew point, and pressure as the most significant factors can be plausibly explained as follows: under the influence of a low-pressure system and associated winter storms, the reduction in atmospheric moisture lowered the dew point, and this—combined with the extreme cold—triggered an unanticipated surge in electricity demand as residents sought to heat their homes. Had a similar winter storm occurred in 2017 or 2018, prior to the coronavirus disease 2019 (COVID-19) pandemic, it is conceivable that Texas might not have experienced such a widespread blackout. However, during the pandemic, students were attending remote classes from home while operating heating systems, and remote-working employees were likewise spending more time in their homes, collectively driving unprecedented spikes in electricity consumption.

This unexpected demand pattern appears to have been a critical factor in causing the large-scale power outages in February 2021. Despite the Electric Reliability Council of Texas (ERCOT)'s forecasts being based on historical data from 2020 or earlier, the unprecedented behavioral changes induced by COVID-19 likely rendered such forecasts less accurate and inadequate for managing real-time demand during the crisis. It is important to note that ERCOT functions as an independent system operator, orchestrating the intricate and continuous management of electrical power distribution to approximately 24 million consumers within the state of Texas. This operational jurisdiction encompasses nearly 90 percent of the state's total electrical load, underscoring ERCOT's pivotal role in sustaining Texas's energy infrastructure and ensuring the stability of its power grid (Electric Reliability Council of Texas, 2025).

In the XGBoost analysis, categorical variables such as State, Weather Condition, and Wind Direction did not emerge as highly influential predictors. While it is plausible that the state of Texas, being predominantly affected, may exert some degree of influence on outage prediction, this influence appears to be comparatively weaker than that of variables such as population, dew point, and atmospheric pressure.

5.5 Partial dependence plots

Figure 9 shows the results of partial dependence plots (PDPs) for the six most significant features chosen from the top ten variables ranked by permutation importance via the XGBoost model. Although several nonparametric models (e.g., LightGBM) were explored in this study, the interpretation centers on XGBoost due to its superior predictive performance based on the evaluation metrics. In a PDP, "high partial dependence" indicates that changes in the value of a specific feature (like dew point) lead to significant shifts in the model's predictions. Conversely, a "low partial dependence" suggests that the model's prediction hardly adjusts even as the value of that feature changes.

As the adjusted population grows, the predicted value also rises steadily. A sharp increase is seen with approximately 60,000 people. The anticipated value keeps rising gradually, and there are areas where the graph levels off. Approximately 320,000 people, a significant increase takes place. Following the sudden rise, the anticipated value stays elevated or exhibits a minor upward movement. In general, as the population grows, the expected value (the number of power outages) usually rises. Remarkably, around 60,000 people and 300,000 people, there are substantial increases in the projected value. These incremental increases indicate that, within certain limits, population size has a notable impact on the model's forecasts.

For the area, the predicted values are generally flat or show slight fluctuations up and down. In the range of 500–600 mi^2, the anticipated values indicate a small decline. The graph remains level, but at approximately 800 mi^2, it abruptly escalates. From the 800 mi^2 + range, the predicted values start to increase noticeably. Although there are occasional fluctuations (sharp rises and falls), in general, as the area increases, the predicted values clearly rise. Around 1,300 mi^2, the predicted values reach their peak. In other words, larger areas can be seen as the ranges where the model's predictions show greater influence on the number of power outages.

For the atmospheric pressure, the predicted values remain mostly steady around 5,900–6,100 households, showing only a slight increase in 27.3–28.4 inHg range. In this range, atmospheric pressure has almost no influence on the predictions. The predicted values gradually decrease in 28.4–29.0 inHg range. Around 29.0 inHg, the downward trend becomes more pronounced. The predicted values fluctuate irregularly and then begin to drop rapidly in 29.0–29.9 inHg range. Although there are oscillations (ups and downs) in the middle, the overall trend is downward. The predicted values decrease sharply,

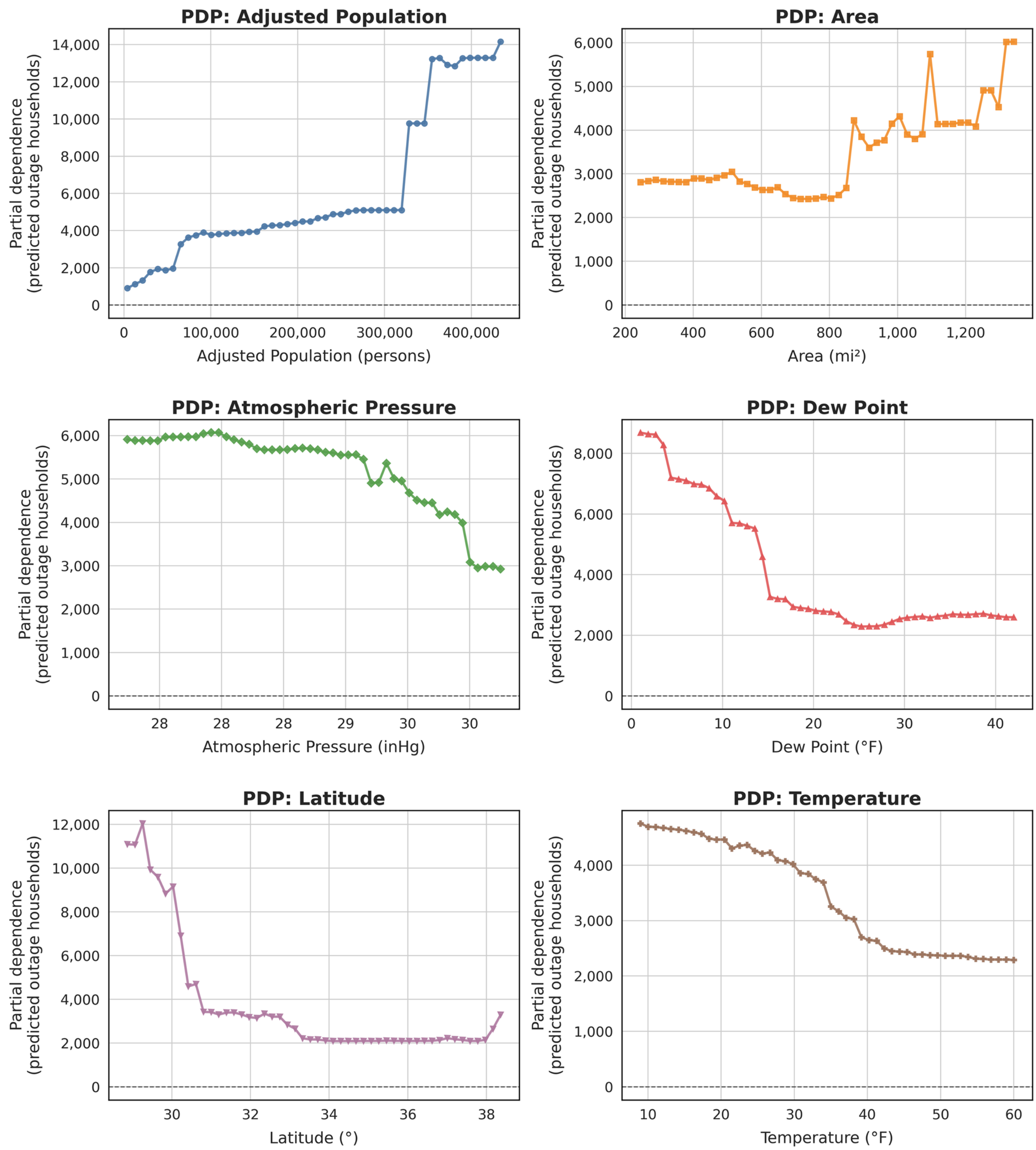

Fig. 9 Results of PDPs for XGBoost

falling into the 3,200–3,000 households range between 29.9 inHg and 30.2 inHg for the pressure. Large changes occur in this range. As atmospheric pressure increases, the model's predicted values decrease significantly. In particular, at 29.9 inHg and above, atmospheric pressure plays a very strong role in reducing the predicted values.

When the dew point is low, the predicted value starts at 9,000 households and decreases sharply. In particular, around 10–15 °F of the dew point, the predicted value drops significantly. The predicted value falls rapidly into approximately 3,000 households, and the graph takes on a step-like shape in 15–20 °F range. After this, the rate of decline slows down. The predicted value remains nearly constant between 2,500 households and 3,000 households in 20–45 °F range of the dew point. In this range, the dew point shows little influence on the predictions. When the dew point is low (drier conditions), the predicted values are very high, but once the dew point rises above about 10–15 °F, the predictions decrease rapidly. As the dew point rises further (more humid conditions), the predicted values continue to decrease, but beyond a certain level, the influence becomes minimal.

The predicted values start very high, at around 10,000–12,000 households in low latitude (around 29–30°) As it passes 30°, the predicted values drop sharply. The predicted values fall rapidly, decreasing to about 3,000–4,000 households in 30–31° range of the latitudes. In this range, the model undergoes very significant changes in its predictions. The partial dependence values remain relatively flat, meaning that even as latitude increases further, the predictions do not change much in 31–37° range of latitude. They stay roughly in the range of 2,000–3,000 households, with very little variation. There seems to be a slight upward trend in the predicted values, but no dramatic changes occur beyond 37° of the latitude. At lower latitudes (further south), the model's predicted values (the number of power outages) are extremely high. Around the low 30 s, the predictions drop sharply and afterwards remain stable. In other words, the 30–31° range is a critical turning point for the power outages prediction. In relatively higher-latitude regions, the model's predicted values are steady and comparatively low.

As the temperature increases, the predicted values gradually decrease in 10–35°F range of the temperature. Up to around 35 °F, the values show a steady downward trend. In 35–40°F range, the predicted values drop most sharply, forming a "steep slope." Once the temperature surpasses 35 °F, the model significantly lowers its predictions. After dropping to around the 2,500 households of partial dependence, the predicted values flatten out (plateau) in 40–60°F range. Despite the ongoing increase in temperature, the model's forecasts reveal minimal alterations. At low temperatures (10–35°F), the predicted values of the model (the number of power outages) are elevated. Between 35–40°F, the forecasts decline significantly. Above 40 °F, the predictions remain low and further increases in temperature have little effect. In other words, the occurrence of power outages rises in colder regions; however, once the temperature exceeds 35–40°F, it has minimal effect on the model's predictions.

5.6 Limitations of this study and future work

Data on power outages caused by winter storms is limited when compared to research on other natural hazards and their impact on the power infrastructure, such as hurricanes. Especially in the southern region, winter storms occur with extremely low probability, leading to inadequate preparedness for such events in this area. Consequently, unprecedented large-scale power outages occurred in the southern region. Additionally, the data extracted for this study only included winter storm Uri affecting the southern U.S., which includes a 2- to 3-h interval data for two weeks, and obtaining publicly available data on power outages for baselines of predicting power outages during winter storms was challenging. However, if the data is sufficiently secured, the predictive model's performance will show improved results.

This study developed an outage prediction model using data from power outages caused by winter storm Uri in the southern region of the U.S. A comparative experiment with the outage prediction model for ice/snowstorms developed by Cerrai et al. (2020), which was centered on the state of Connecticut, would make it possible to more clearly ascertain the enhanced applicability of the proposed approach in the southern region. However, due to the closed nature of the data, a direct comparison with their work was challenging. First, their dataset was obtained directly from utility companies, making data acquisition difficult. Second, their data were recorded at the town level—that is, at a higher spatial resolution than the county-level data used in the present model—which would hinder direct comparison. Nevertheless, conducting future research under conditions that closely match those of existing models developed for northern regions would allow for a more definitive demonstration of the improved applicability of the proposed model in southern settings.

As discussed earlier, population and dew point variables exhibit considerable explanatory potential. Nevertheless, augmenting the model with infrastructure-related data could augment its predictive performance. Furthermore, with the accumulation of additional outage data due to other winter storms in the southern US, the model is anticipated to be refined, leading to heightened precision and robustness.

In addition, if the proposed model were applied to power-dispatch optimization, electricity should be allocated preferentially to regions with a high risk of outages. This is primarily because the large-scale blackout during the 2021 winter storm Uri was largely driven by an imbalance between electricity demand and supply. In the event of another severe winter storm, it would be prudent to maintain more generous reserve capacity, taking the 2021 incident as a reference. Investigations into the level of reserve capacity necessary to prevent large-scale blackouts such as the one in 2021 would also be highly valuable. From a policy-making standpoint, the current model appears to have certain limitations in enabling proactive measures, which constrain its utility for direct policy formulation. Since the model predicts outages using only current data, further research is required to establish response manuals that employ data from one, two, or even three days in advance, thereby enabling anticipatory action rather than relying solely on present conditions. Although further research is undoubtedly warranted, it is considered that information on current outage levels can still serve as a useful basis for policy makers and emergency response teams in formulating rapid-response strategies.

In this study, data augmentation was applied to a single event, Winter Storm Uri, but limitations remain regarding the generalizability of the findings. Future research will seek to address and mitigate these constraints by employing synthetic grid-based simulations or incorporating multi-event datasets. To enable meaningful synthetic grid-based simulation, utility companies and electricity facilities must share relevant data for the public interest, and continuous consensus-building among stakeholders is essential.

For example, such simulations require detailed information like "bus" data within the power grid, which play key roles in power distribution and voltage management. Bus data represent actual locations where power supply and demand are determined. Since misuse of these datasets can pose risks to public safety, careful collaboration and agreement among municipalities, utilities, and power providers are necessary for accurate simulations.

Furthermore, by considering multi-event datasets in future work, it will be possible to overcome the limitations in generalizability that stem from analyzing a single event. For instance, developing outage prediction models based on multiple winter storm events in southern regions including Texas, or building and validating models for hurricanes as well, will lead to more generalizable and robust predictive frameworks. However, constructing such models requires substantial quantitative and qualitative big data. For hurricane-specific outage prediction models, variables highlighted in previous research, such as soil type and leaf area index, must be carefully considered and incorporated.

5.7 Implementation strategies for utilities

Utility operators can implement the proposed model by prioritizing the collection and integration of highly influential variables, such as population, dew point, and pressure, into their demand forecasting processes to optimize operational planning and enhance predictive accuracy. Utility operators may rapidly deploy either the simplest model or a model utilizing all available variables to predict and help prevent or minimize outage-related damages. Nonetheless, additional validation and assessment of model generalizability are required, especially by integrating outage caused by multi-event datasets or synthetic grid system simulations.

This research was conducted in the southern United States, where winter storms are uncommon and storm-induced outages occur infrequently. This context should be remembered. In areas prone to frequent and intense winter storms, the findings of this study should be viewed as supplemental, and further factors like leaf area index (indicating vegetation) need to be included. damage electrical infrastructure, causing power outages.

Key factors in forecasting power outages due to hurricanes are wind speed, leaf area index, and soil type, as these storms often bring substantial precipitation. The water retention characteristics of soil types can affect the stability of utility poles; when the soil is saturated and soft, poles are more likely to lean or topple in high winds, resulting in grid instability. If utility operators develop event-specific optimized models or outage prediction models based on multi-event data, and further integrate these with synthetic grid-based simulations, they can verify, enhance generalizability, and ultimately improve the robustness of outage prediction systems.

These recommendations are anticipated to be relevant in areas outside of the United States, due to comparable mechanisms in power systems worldwide. However, additional confirmation would be required for those regions. Additionally, utility operators may team up with electrical engineers to obtain reliable outage datasets that consider multi-event situations or to create sophisticated models through synthetic power grid simulations. According to this study, operators should concentrate on obtaining real-time or big data for the key variables and employ these strong models to assist decision-makers in reducing losses by forecasting and preparing for outages beforehand.

6 Conclusions

Winter storm Uri landed in February 2021, causing extensive power outages across the southern regions of the United States. A power outage prediction model is necessary to prevent such occurrences in the future. For this reason, the purpose of this study is to find an optimal power outage prediction model among four non-parametric machine learning regressions, in comparison with traditional statistical models such as Ridge and Lasso, and to identify the most influential variables. The results are as follows.

The dataset was split into 20% for the test set and 80% for the training set. To enhance generalization and achieve a more robust model, Gaussian data augmentation was applied to the training set. For model optimization, Bayesian optimization was employed, with the objective function defined as the Root Mean Squared Error (RMSE). Model evaluation was conducted using tenfold cross-validation, based on four metrics: R^2, RMSE, Mean Absolute Error (MAE), and Median Absolute Percentage Error (MdAPE). The results indicated that XGBoost demonstrated the best performance, with corresponding metric values of $R^2 = 0.99$, RMSE = 849.66 households, MAE = 375.09 households, and MdAPE = 98.46%. When evaluated on the test set, XGBoost again achieved the most favorable results, with $R^2 = 0.92$, RMSE = 4,991.90 households, MAE = 1,192.10 households, and MdAPE = 100.00%. These test results were found to be consistent with the outcomes of the tenfold cross-validation, thereby confirming the reliability and robustness of the model's performance. According to the results of permutation importance, dew point, population (adjusted), and pressure consistently ranked within the top three most important variables across all models, except for Ridge and Lasso. This suggests that the dew point is crucial in explaining power generation disruptions due to winterization. Moreover, pressure, organically linked to the dew point, indirectly demonstrates power outages. Additionally, considering the imbalance in electricity supply due to excessive demand, population (adjusted) indirectly explains power outage predictions well.

In this work, six fundamental models for power outage prediction during winter storms in the southern region have been developed. Although all models, except for Ridge and Lasso, can sufficiently serve as a basic prediction model for winter storm-induced outages, XGBoost is selected as the most optimal model for its efficiency in handling different datasets based on cross-validation results and overall accuracy metrics. Additionally, by analyzing all six models, the key factors that accurately predict power outages during winter storms were determined. Moreover, considering there is insufficient data regarding winter storms in the southern region, it is crucial to continue to collect comprehensive datasets on future winter storms in this region, thus improving the training data for the models produced in this work and, eventually, resulting in a more robust predictive model.

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Authors' contributions
Jangjae Lee: Methodology, Data curation, Formal Analysis, Resources, Software, Validation, Visualization, Writing – original draft; Zhe Zhang: Conceptualization, Resources, Validation, Methodology, Writing – review & editing; Stephanie G. Paal: Conceptualization, Methodology, Visualization, Resources, Supervision, Writing – review & editing.

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Data availability
The corresponding author will provide all necessary data for this study upon request.

Declarations

Competing interests
The authors declare that they have no competing interests.

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